PROBLEM 12

Conservative Force and Potential III

Original statement · Problems.pdf, p. 9

Oscillation Period

A particle moving under a conservative force oscillates between x_1 and x_2. Show that the period of oscillation is:

$$\tau = 2\int_{x_1}^{x_2} \sqrt{\frac{m}{2[V(x_2) - V(x)]}}\, dx$$

where $V(x)$ is the potential energy function. In particular, if $V(x) = \frac{1}{2}m\omega_0^2(x^2 - bx^4)$, show that the period for oscillations of amplitude a is:

$$\tau = \frac{2}{\omega_0}\int_{-a}^{a} \frac{dx}{\sqrt{a^2 - x^2}\sqrt{1 - b(a^2 + x^2)}}$$

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m undergoes one-dimensional bounded periodic motion under a conservative potential $V(x)$ between two consecutive turning points x_1 and x_2.
- **General Energy Law:** Total mechanical energy $E = T + V(x)$ is conserved.
- **Turning Point Conditions:** At the extremities $x = x_1$ and $x = x_2$, the particle comes to rest:

$$v(x_1) = v(x_2) = 0 \implies E = V(x_1) = V(x_2)$$

- **Specific Potential Model:**

$$V(x) = \frac{1}{2} m \omega_0^2 (x^2 - b x^4)$$

where ω_0 is the natural small-amplitude frequency $[\mathrm{rad/s}]$, and b is the non-linear anharmonicity parameter $[\mathrm{m}^{-2}]$. Amplitude of oscillation is a ($x_1 = -a, x_2 = +a$).

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Kinematic Relation:**

$$v = \frac{dx}{dt} \implies dt = \frac{dx}{v}$$

- **Kinetic Energy Isolation:**

$$T = \frac{1}{2}mv^2 = E - V(x) \implies v^2 = \frac{2}{m}[E - V(x)]$$

$$v = \pm\sqrt{\frac{2}{m}[E - V(x)]}$$

PHASE 3

Step-by-Step Mathematical Deduction

1. General Derivation of the Oscillation Period τ

Pedagogical Justification: Because the force is conservative, the equations of motion are invariant under time reversal ($t \to -t$). The time taken to travel forward from x_1 to x_2 with positive velocity $v = +\sqrt{\frac{2}{m}[E - V(x)]}$ is identical to the return time from x_2 to x_1 with negative velocity $v = -\sqrt{\frac{2}{m}[E - V(x)]}$. Therefore, the total period τ is exactly twice the half-period transit integral.

The differential time increment along the forward path is:

$$dt = \frac{dx}{\sqrt{\frac{2}{m}[E - V(x)]}} = \sqrt{\frac{m}{2[E - V(x)]}}\,dx$$

Integrating from $x = x_1$ to $x = x_2$, the half-period is:

$$\frac{\tau}{2} = \int_{x_1}^{x_2} \sqrt{\frac{m}{2[E - V(x)]}}\,dx$$

Since x_2 is a turning point, the total energy is precisely $E = V(x_2)$. Substituting $E = V(x_2)$:

$$\tau = 2\int_{x_1}^{x_2} \sqrt{\frac{m}{2[V(x_2)-V(x)]}}\,dx$$

This proves the general theorem. ■

2. Application to the Anharmonic Potential $V(x) = \frac{1}{2}m\omega_0^2(x^2 - bx^4)$

> **Pedagogical Justification:** For symmetric oscillations of amplitude a, the turning points are $x_1 = -a$ and $x_2 = +a$. We evaluate the energy difference $V(a) - V(x)$ algebraically and factor out the singular harmonic core $(a^2 - x^2)$.

Evaluate $V(a)$:

$$V(a) = \frac{1}{2}m\omega_0^2(a^2 - ba^4)$$

Form the difference $V(a) - V(x)$:

$$V(a) - V(x) = \frac{1}{2}m\omega_0^2(a^2 - ba^4) - \frac{1}{2}m\omega_0^2(x^2 - bx^4)$$

Factor out $\frac{1}{2}m\omega_0^2$:

$$V(a) - V(x) = \frac{1}{2}m\omega_0^2\left[(a^2 - x^2) - b(a^4 - x^4)\right]$$

Apply the algebraic identity for difference of squares:

$$a^4 - x^4 = (a^2 - x^2)(a^2 + x^2)$$

Substitute this identity into the bracketed term:

$$V(a) - V(x) = \frac{1}{2} m\omega_0^2 \left[(a^2 - x^2) - b(a^2 - x^2)(a^2 + x^2) \right]$$

Factor out the common term $(a^2 - x^2)$:

$$V(a) - V(x) = \frac{1}{2} m\omega_0^2 (a^2 - x^2) \left[1 - b(a^2 + x^2) \right]$$

3. Assembly of the Period Integral

Substitute this factorized expression into the denominator of the general period formula:

$$2[V(a) - V(x)] = 2 \left(\frac{1}{2} m\omega_0^2 (a^2 - x^2) \left[1 - b(a^2 + x^2) \right] \right) = m\omega_0^2 (a^2 - x^2) \left[1 - b(a^2 + x^2) \right]$$

Now take the square root of the ratio:

$$\sqrt{\frac{m}{2[V(a) - V(x)]}} = \sqrt{\frac{m}{m\omega_0^2 (a^2 - x^2)[1 - b(a^2 + x^2)]}} = \frac{1}{\omega_0 \sqrt{a^2 - x^2} \sqrt{1 - b(a^2 + x^2)}}$$

Substitute this integrand into the period equation with limits from $-a$ to $+a$:

$$\tau = 2 \int_{-a}^{a} \frac{dx}{\omega_0 \sqrt{a^2 - x^2} \sqrt{1 - b(a^2 + x^2)}}$$

Factor out the constant ω_0:

$$\tau = \frac{2}{\omega_0} \int_{-a}^{a} \frac{dx}{\sqrt{a^2 - x^2} \sqrt{1 - b(a^2 + x^2)}}$$

This completes the exact derivation. ■

PHASE 4

Physical Interpretation & Dimensional Verification

- **Harmonic Limit ($b \to 0$):**

When $b = 0$, the potential reduces to the simple harmonic oscillator $V(x) = \frac{1}{2}m\omega_0^2 x^2$.

The integral becomes:

$$\tau = \frac{2}{\omega_0}\int_{-a}^{a} \frac{dx}{\sqrt{a^2 - x^2}}$$

Using the trigonometric substitution $x = a\sin\theta$, $dx = a\cos\theta\, d\theta$, with limits $\theta \in [-\pi/2, \pi/2]$:

$$\int_{-a}^{a} \frac{dx}{\sqrt{a^2 - x^2}} = \int_{-\pi/2}^{\pi/2} \frac{a\cos\theta\, d\theta}{a\cos\theta} = \int_{-\pi/2}^{\pi/2} d\theta = [\theta]_{-\pi/2}^{\pi/2} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

Multiplying by $\frac{2}{\omega_0}$:

$$\tau = \frac{2\pi}{\omega_0}$$

which recovers the classical harmonic period $\tau_0 = 2\pi/\omega_0$, completely independent of the amplitude a (isochronism).

- **Anharmonicity ($b > 0$):**

For a softening spring ($b > 0$), the factor $\sqrt{1 - b(a^2 + x^2)} < 1$, which makes the integrand strictly larger than in the harmonic case, proving that $\tau > \tau_0$ and that the period grows with amplitude a.

- **Dimensional check:** $[b] = \mathrm{m}^{-2}$, so $b(a^2 + x^2)$ is dimensionless; $\int dx/\sqrt{a^2 - x^2}$ is dimensionless and $[\tau] = [1/\omega_0] = \mathrm{s}$.

- **Small anharmonicity:** with $x = a\sin\varphi$, $dx = a\cos\varphi\,d\varphi$ the period reads $\tau = \dfrac{2}{\omega_0}\displaystyle\int_{-\pi/2}^{\pi/2}\frac{d\varphi}{\sqrt{1 - ba^2(1+\sin^2\varphi)}}$. Expanding to first order in ba^2 and using $\langle\sin^2\varphi\rangle = 1/2$ gives $\tau \approx \dfrac{2\pi}{\omega_0}\left(1 + \dfrac{3}{4}ba^2\right)$.
- **Admissible amplitudes:** V has its maximum at $x^2 = 1/(2b)$, so bounded oscillations require $a^2 < 1/(2b)$; as $a^2 \to 1/(2b)$ the factor $1 - b(a^2 + x^2)$ vanishes at the turning points and $\tau \to \infty$.
- **Numerical check ($\omega_0 = 1$, $b = 1$; quadrature against RK4 of $\ddot{x} = -x + 2bx^3$, step 10^{-4}):**
- $a = 0.1$: quadrature 6.3309, RK4 6.3312, first-order estimate 6.3303.
- $a = 0.3$: quadrature 6.7592, RK4 6.7592, estimate 6.7073.
- $a = 0.5$: quadrature 8.0086, RK4 8.0088, estimate 7.4613.
- $a = 0.7$ (close to $1/\sqrt{2} = 0.7071$): quadrature 16.947, RK4 16.948; the period grows without bound as $a \to 0.7071$ (36.6 at $a = 0.7071$ in the quadrature).

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